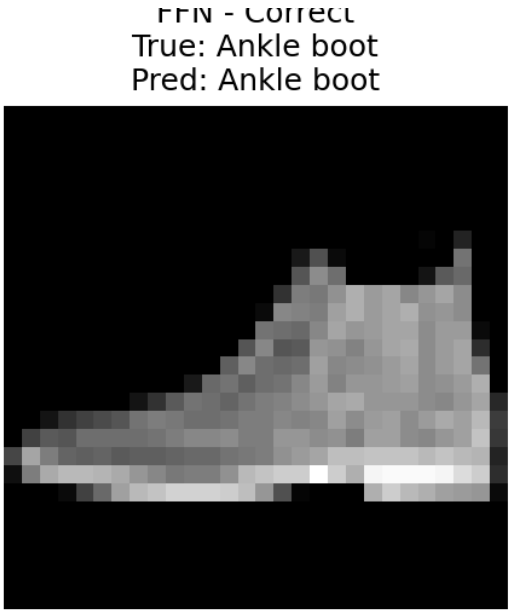
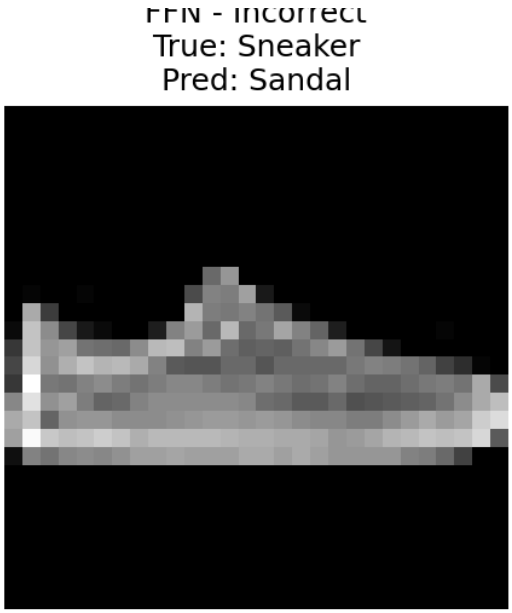
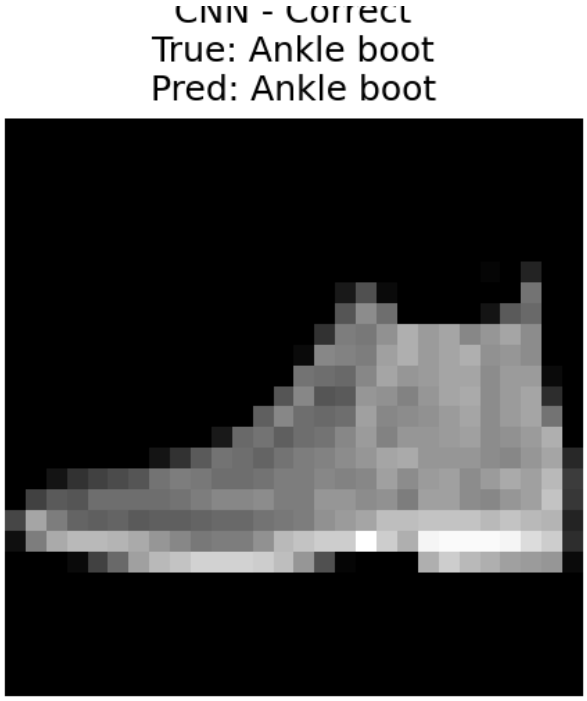
1.

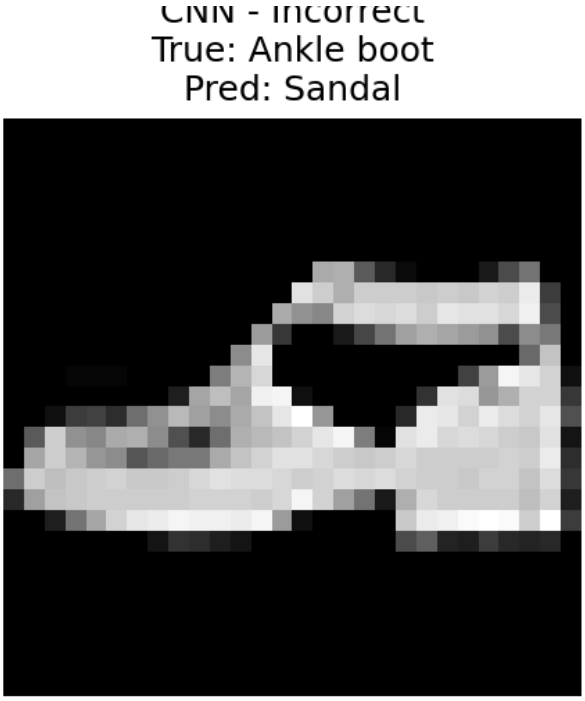
FFN


FFN - Correct
True: Ankle boot
Pred: Ankle boot


FFN - Incorrect
True: Sneaker
Pred: Sandal

CNN

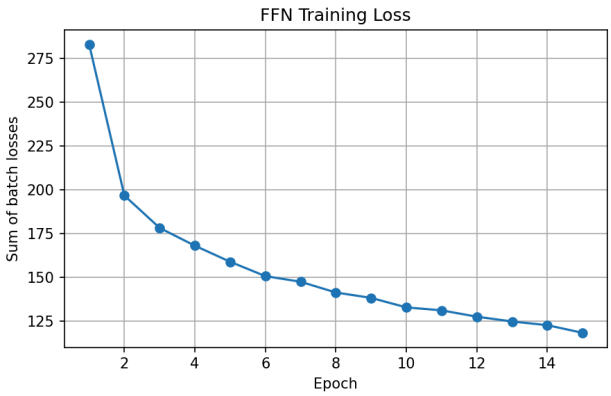

CNN - Correct
True: Ankle boot
Pred: Ankle boot

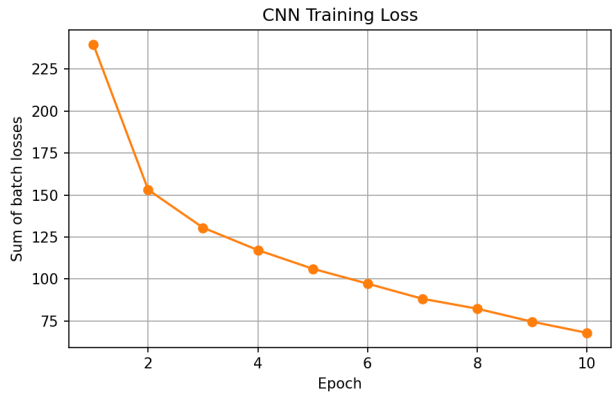

CNN - Incorrect
True: Ankle boot
Pred: Sandal

2.

Training loss (sum of batch cross entropy) per epoch.

FFN: 15 epochs. CNN: 10 epochs.





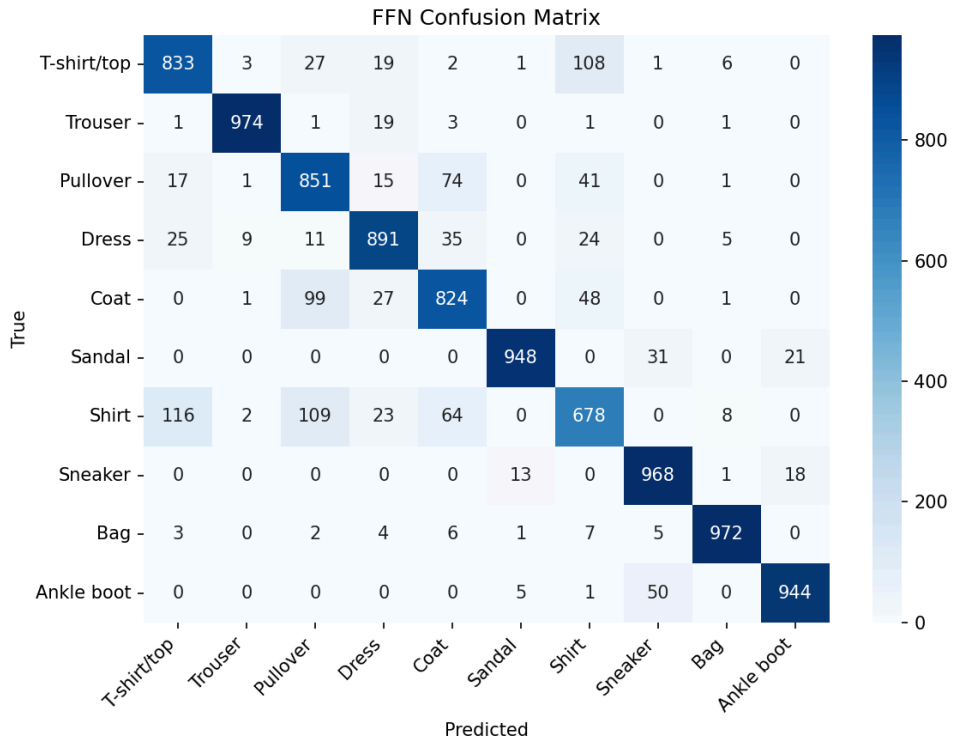
3.

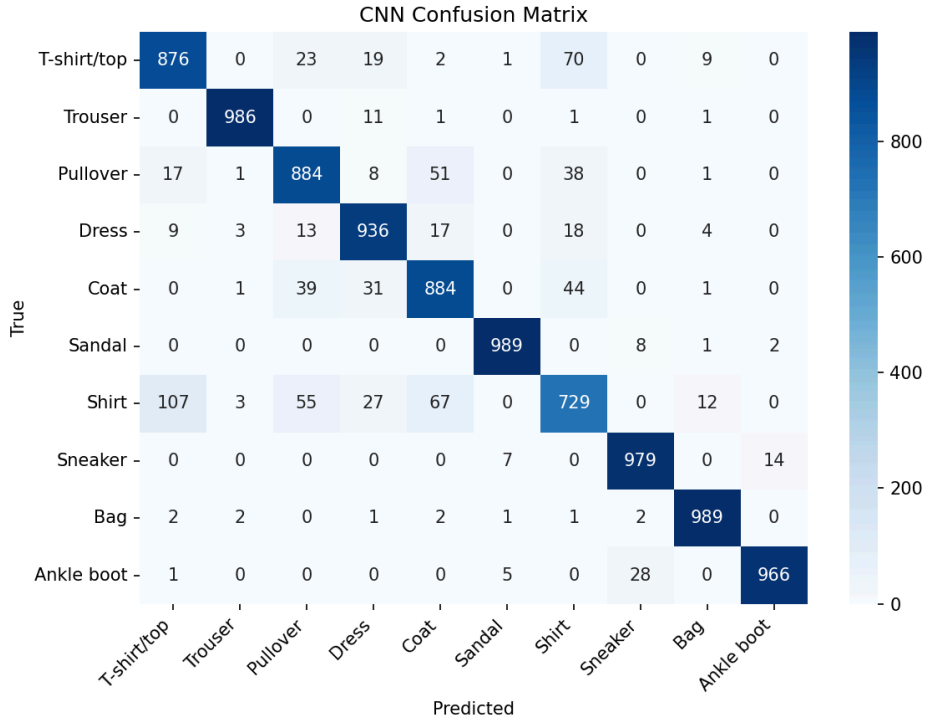
FFN: 242,762

CNN: 421,642

4.

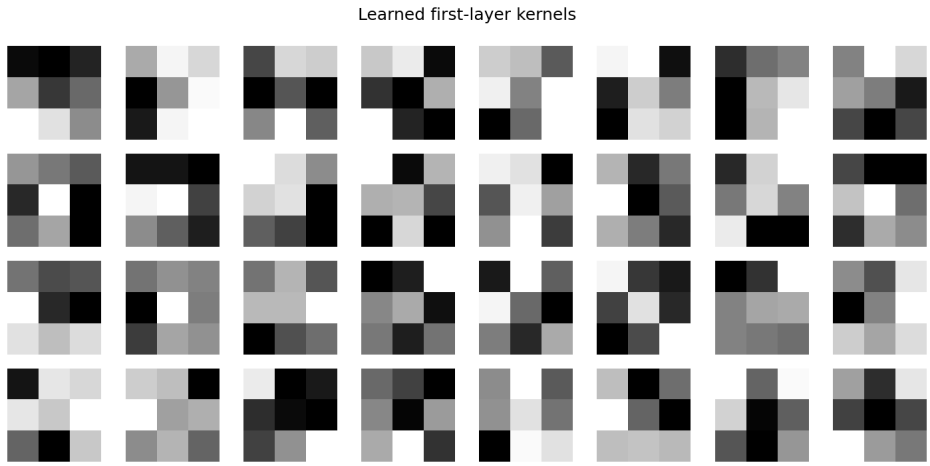
Confusion matrixes on the 10,000 image test set





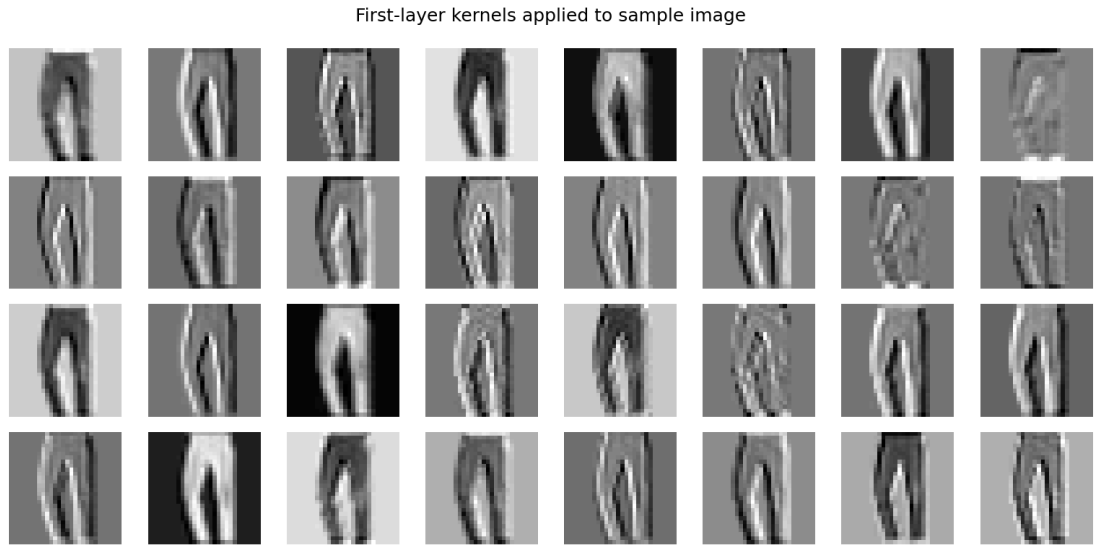
5.

The 32 learned 3×3 kernels from the CNN's first convolutional layer, normalized to [0,1]. Each kernel responds to a different oriented edge or local contrast pattern



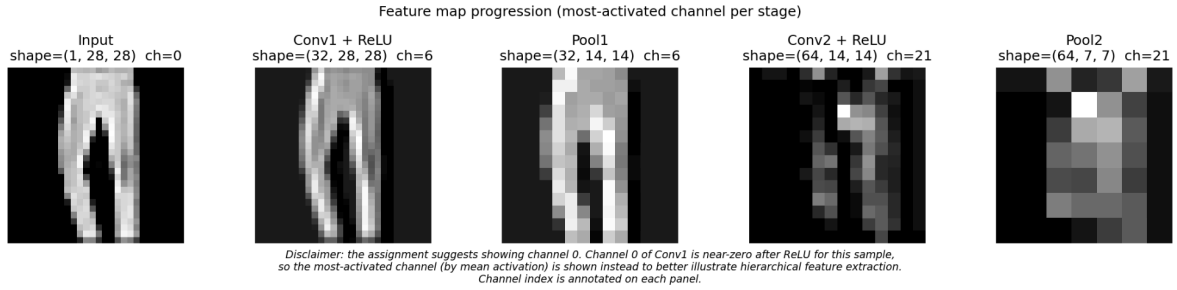
6.

Output of each of the 32 first-layer kernels applied to the sample image. Different kernels emphasize different edges and orientations of the pants



7.

The progression from the input image through every convolutional and pooling layer of the CNN. Channel 0 of Conv1 was near zero after ReLU for this sample, so the most activated channel, is displayed at each stage



8.

A CNN outperforms a fully connected network on images because its convolutional layers exploit the spatial structure of image data. Each kernel is a small set of shared weights slid across every position in the image, so the network learns translation invariant local patterns, like edges and corners, with fewer parameters than an FFN that must learn a separate weight for every connection. Stacked convolutions combined with pooling also build a hierarchy of features, expanding each neuron's receptive field while shrinking spatial resolution and getting rid of irrelevant detail. The fully connected layers at the end only need to reason about the compact, class relevant feature vectors instead of raw pixel arrays. This is consistent with my results: CNN 92.18% v. FFN 88.83%.

9.

The first convolutional layer still resembles the input silhouette but with specific edges and contrast patterns of the pants emphasized, and after Pool1 the spatial resolution is halved so only the strongest low level edge responses survive. The second convolutional layer then composes these primitives into more abstract mid level features

concentrated in a few spatial locations, and Pool2 compresses them into a 7×7 map where each cell summarizes a large block of the original image. This hierarchical refinement from raw pixels to edges to parts to compact class relevant features is what makes classification easy, because by the time the feature map reaches the fully connected head, visually similar clothes already cluster together in feature space so the final linear layers only have to draw simple decision boundaries.